In order to understand those PHP features that are specific to web use cases it's a good idea to look at the fundamentals underpinning the internet itself. Therefore, we'll be looking at the basics of internet application protocols (HTTP, FTP, SMTP), and HTML (Hypertext Markup Language). From there we'll examine common security issues that can arise when dealing with user input (which is never *ever* to be trusted), and in particular when that data has to be passed to other systems (such as databases or passed back to web clients). We'll then close out the chapter by looking at some of the great tools that are out there in Open-Source land that will enable you to build robust, secure, and scalable applications for today's internet. The topics covered in this chapter are

- Internet protocols
- Lifecycle of an HTTP request in PHP
- HTML
- The DOM
- Traversing and modifying the DOM
- Templating
- Input validation and sanitizing

Hypertext Transfer Protocol (HTTP)

Unless you've been living under a rock for the last twenty-five years, you'll have heard of this thing called "the Internet," and likely be more than familiar with its use. You've probably used an application called a web browser (Chrome, Firefox, Edge, Opera… maybe even Internet Explorer if you are of sufficient vintage!) on a device such as a laptop, smartphone, or perhaps even a TV or games console, which magically (or not so magically, depending on the depth of your knowledge) receives beautifully constructed pages of information and functionality from a (usually) remote computer elsewhere in the world. In this scenario, your device is the *client*, and the machine sending you your page is the (web) *server*. This shouldn't be particularly revelatory, especially if you have read Chapter 1, "Getting Started", where we looked at multiple ways of running a web server together with PHP (or indeed, using PHP as a limited-functionality web server). The client and server might even be running on the same machine, which is often the case in a development environment (and also very likely if you're following along at home with the examples in this chapter).